

Telecom Drawing QA Report

Drawing: M8398_DOREEN TOWNSHIP_FC_30052025 (2).pdf

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PASS	FAIL	UNCLEAR	TOTAL
33	9	44	86

Detailed Results

ID	Rule	Verdict	Evidence
R001	Drawings must be correct scale in all the sheets	PASS	Cover sheet (M8398-00) and reference documents sheet (M8398-01) are title/index sheets with no drawing scale required (NTS). Sheet M8398-G1 (Site Specifications) includes a scale bar at the bottom right indicating 1:50 scale on A3, which is a Standard (Preferred) scale per the Optus drawing scale guidelines. No non-standard or 'not preferred' scales were detected across the reviewed sheets.
R002	Check scale with viewport	UNCLEAR	The FC drawing pages provided (M8398-00, M8398-01, M8398-G1) are cover/index and site specification sheets. No viewport scale label (e.g., 'SCALE 1:100') with corresponding viewport properties is visible in the submitted pages to compare against the reference. The site specification sheet shows only a graphical scale bar. Scaled detail/plan drawing pages with viewport scale annotations were not included in the submission, making it impossible to verify scale-to-viewport matching.
R003	FC stamp is correctly mentioned to all sheets?	PASS	All three visible sheets (M8398-00, M8398-01, and M8398-G1) display the 'FOR CONSTRUCTION' stamp in the title block area, confirming the FC stamp is correctly present on all sheets reviewed.
R004	Layers correctly followed	UNCLEAR	The FC drawing pages provided (M8398-00, M8398-01, M8398-G1) are cover/title and site specification sheets containing primarily text and tables, with no actual CAD linework showing existing vs. proposed elements on distinguishable bold/unbold layers. Layer compliance for existing (unbold) vs. proposed (bold) cannot be fully assessed from these pages alone — actual plan, elevation, and detail sheets with drawn site infrastructure would be needed for a definitive verdict.
R005	Site number/Name/Address details correct	UNCLEAR	FC drawing consistently shows Site M8398, DOREEN TOWNSHIP-O, 95 HAZEL GLEN DRIVE, DOREEN VIC 3754, RFNSA No: 3754005 across all pages. Cannot independently verify against RFSNA system — cross-check with RFSNA is required to confirm site number, name, and address details are correct as per the rule requirement.
R006	Add structure owner site ID. (Indara, Amplitel & BAI etc)	PASS	The FC drawing clearly shows the structure owner site ID on the title block of pages M8398-00 and M8398-01: 'INDARA SITE ID: 3501475'. This is consistent with the reference image requirement to include the structure owner (Indara, Amplitel, BAI etc.) site ID. The Site Specifications page (M8398-G1) also references 'EXISTING INDARA ROOFTOP SITE' and mentions 'REFER TO CERT_M8398 DATED 02/04/2025' confirming Indara as the structure owner.
R007	Work authority number is correct	FAIL	The Work Authority Number in the FC drawing is 540268 (shown as 'OPTUS WORK AUTHORITY N° 540268' on the cover sheets), but the reference image (Antenna System tab Scope of Works) shows WA ID 658102. The numbers do not match.

R008	Client logo is correct	PASS	The FC drawing displays the correct Optus logo (teal/cyan colored 'OPTUS' text matching the reference image) on all pages including the title block pages and site specifications sheet. The client is identified as 'OPTUS' and the vendor/contractor is 'ServiceStream' (with ServiceStream logo visible), which is one of the approved vendors (CPS/Servicestream/CAM/Ventia/Nokia).
R009	Drawings listed in cover sheet aligned with previous As-built & proposed scope (Sheet name/Number/title/Rev.)	PASS	The FC cover sheet (M8398-00) lists all drawing sections (GENERAL, ANTENNAS & TRANSMISSION, STRUCTURAL, ELECTRICAL, SHELTER/FITOUT, EME EXCLUSION ZONES, LEASE/LICENCE) with sheet numbers, titles, and revisions. All proposed/new drawings in the current FC column (version 11, dated 02.04.25) are marked as REV 'A', consistent with the rule requirement. Previous as-built revisions are carried forward in earlier version columns. The drawing structure and revision marking convention aligns with the reference image pattern.
R010	MERC signage is shown in the FC pack as per the latest Optus OSD requirements - Site specific for typical ground site - OSD-171-1, Rooftop/watertower OSD-171-2 & Site signage legend and notes OSD-171-3.	PASS	Site is identified as a Rooftop site (REF LOCATION: 4. ROOFTOP on M8398-G1). The Reference Documents sheet (M8398-01) correctly includes OSD-171-2 (SITE SIGNAGE TYPICAL ROOFTOP SITE – PLAN, Rev B) and OSD-171-3 (SITE SIGNAGE LEGEND AND NOTES, Rev B). OSD-171-1 (ground site) is correctly excluded. The site specifications sheet also confirms signage to be in accordance with OSD-171-2 and OSD-171-3.
R011	Add reference documents - Add Pole & Mount Structural cert which dates matches with provided cert. Also check with FC issue date	PASS	The Reference Documents section on sheet M8398-01 lists CERT_M8398 as 'SUPPLIER STRUCTURAL DESIGN CERTIFICATE – ROOFTOP FOR INDARA INFRASTRUCTURE & ENGINEERING DATED 02-04-2025'. This date (02/04/2025) matches the latest FC issue date of 02.04.25 shown on the drawing package version history, satisfying the rule that the pole/mount structural certificate date must match both the provided certificate and the FC issue date.
R012	Service stream site only - Add "SAED_DA/DC/L3/LO CONDITIONS" in reference documents (Refer snip 8 in next sheet for details)	FAIL	The FC drawing (M8398-01) Reference Documents section does not include 'SAED_DA/DC/L3/LO CONDITIONS'. This is a ServiceStream site (confirmed by ServiceStream logo and distribution entry), and the required entry is missing from the reference documents list, whereas the reference image clearly shows it should be present and highlighted.
R013	Distribution list: (Optus & either CPS or Servicestrem) OPTUS : SUHAIB OBAID CPS TECH : BRETT THOMSON Service stream: SAMINA TABASSUM	PASS	The FC drawing distribution list shows OPTUS: SUHAIB OBAID and SERVICE STREAM: SAMINA TABASSUM, which matches the required rule (Optus & either CPS or ServiceStream). The vendor is ServiceStream (confirmed by logo on drawing), and the correct ServiceStream representative SAMINA TABASSUM is listed. This is consistent across all drawing pages checked (M8398-00, M8398-01, and M8398-G1).
R014	Revision numbers correct (Eg."A" issue)	PASS	All FC drawing sheets show Revision 'A' in the latest issue column (version 11, dated 02.04.25). The site specifications sheet M8398-G1 explicitly displays 'A' in the revision block of the title block, matching the reference image requirement. Cover sheet and reference document sheet also confirm revision 'A' in their respective title blocks.
R015	Site address details correct?	PASS	Site address details are correctly populated: DOREEN TOWNSHIP, 95 HAZEL GLEN DRIVE, DOREEN VIC 3754, Optus Site M8398, RFNSA No: 3754005. Coordinates (Easting: 334 713, Northing: 5 838 372, Latitude: -37.58608°, Longitude: 145.12796°) are sourced from RFNSA. Details are consistent across all drawing pages including cover sheet and site specifications page.
R016	Check "Existing structure sections TITLE" like height, model	PASS	The FC drawing (M8398) shows 'EXISTING INDARA ROOFTOP SITE' with an existing antenna mounting pole with turret mount on 8.65m high building parapet wall, with owner identified as INDARA. The structure type (rooftop) and owner name (Indara) are consistent throughout the drawing pages. No Telstra-to-Optus ownership transfer note is applicable for this site, and no mismatch in structure type or owner name was identified.

R017	Point 1 "Existing structure sections" Mention existing tower with heights, model & Owner. Like below format; EXISTING INDARA 54.76m HIGH ROAM RT84 SELF SUPPORTING LATTICE TOWER	FAIL	The FC drawing (M8398-G1) section heading reads 'EXISTING INDARA ROOFTOP SITE' but does not include the required format stating the structure height, model, and owner together (e.g., 'EXISTING INDARA Xm HIGH [MODEL] ROOFTOP POLE'). The 8.65m referenced in point 1 refers to the building parapet wall height, not the structure description in the required format. The model of the mounting pole/turret is not mentioned in the existing structure section as required.
R018	Structural adequacy of pole & foundation (Point 2)	PASS	The FC drawing references structural certificate CERT_M8398 (Supplier Structural Design Certificate – Rooftop) for Indara Infrastructure & Engineering dated 02-04-2025, listed in the Reference Documents (M8398-01). The Site Specifications page (M8398-G1) confirms under Point 3 that structural adequacy of existing antennas and RRU mounts has been confirmed and certified by AllTelcos, with reference to CERT_M8398 dated 02/04/2025. The certifier name and date are present, and the structure is confirmed as passed.
R019	Check callout for Panel/RRU/AAU to be installed on existing or new mount (Point 3)	PASS	The site specifications (M8398-G1) clearly state in Point 2 that new panel antennas are to be installed on existing turret on existing mounting pole, and new RRUs on existing RRU mount. Point 3 confirms structural adequacy has been certified by Allelecoc with reference to CERT_M8398 dated 02/04/2025. No new mounts are proposed, so no new structural drawing number is required. The drawing register also lists antenna and RRU mount detail sheets (M8398-S4 to S11) and the VPL structural certification (VPL-SC-M8398) as reference documents, consistent with the existing mount scenario.
R020	Structural adequacy of mount (Point 4)	PASS	Point 3 of the Existing Indara Rooftop Site notes structural adequacy of existing antennas and RRU mounts has been confirmed and certified by ALLTELECOS, referencing CERT_M8398 dated 02/04/2025. The certificate is also listed as a reference document (CERT_M8398 - Supplier Structural Design Certificate – Rooftop, Indara Infrastructure & Engineering, dated 02-04-2025). The mount is confirmed as structurally adequate (passed), and no strengthening is required.
R021	Is the antenna maintenance access by EWP only or access step pegs with Lad Saf?. If Lad-saf are present at existing, please check which is certified or not. (Point 5)	FAIL	The FC drawing (M8398-G1, point 4 under Existing Indara Rooftop Site) states antenna maintenance access is via 'LADDER AND STEP PEGS WITH FALL ARREST SYSTEM,' and construction access notes reference a Lad-Saf cable on site. However, the drawing does not confirm whether the Lad-Saf/fall arrest system is certified or not, which is required per the rule when Lad-Saf is present at the existing site.
R022	Update the Equipment Shelter as per the As-Built (AB). These are mostly in the unbold layer unless an update is required.	PASS	The Equipment Shelter callout in the FC drawing (M8398-G1) correctly describes the existing OPTUS PHASE 8 (3m x 2.5m) sandwich panel shelter supported on steel frame, referencing M8398-F1 for further details. No footing structural adequacy details are included, which is correct for an existing/upgrade site as shown in the reference image where such items are marked as 'not required, which is previous scope'.
R023	Update the Transmission as per the As-Built (AB). These are mostly in the unbold layer unless an update is required.	PASS	The FC drawing (M8398-G1) correctly states 'THIS SITE IS LINKED TO THE NETWORK VIA EXISTING FIBRE' in the Transmission section, presented in the unbold layer with no update markup required. No active Optus dish is referenced anywhere in the FC drawing pages, consistent with a fibre-connected site. The transmission type is correctly documented as per the As-Built requirement.
R024	Site hazards updated as per site specific requirements. (Refer previous AB & SDV photos)	PASS	The EXISTING SITE HAZARDS section on drawing M8398-G1 includes 9 site-specific hazards: EME Transmitting Antennas, Manual Handling, Working at Heights, Slip/Trip and Falls, Electrical Hazards, Weather/Lightning, Sun Exposure, Unprotected Roof Edges, and Traffic Management. These are appropriate for this urban rooftop site (95 Hazel Glen Drive, Doreen), matching the reference image requirements including Sun Exposure, Unprotected Roof Edges (rooftop with ladder/step peg access), and Traffic Management as noted in the construction access notes.

R025	Site signage needs to be updated as per OSD-171, PRD, SDV photos & FORM A/B	PASS	The FC drawing (M8398-G1) correctly references OSD-171-2 (Rooftop Site) and OSD-171-3 (EME Signage/Legend) in the Site Signage section, which is appropriate since the site is identified as a rooftop site (REF LOCATION: 4. ROOFTOP). The reference image confirms rooftop sites should use OSD-171-2. OSD-171-3 is also correctly included. The reference documents sheet (M8398-01) also lists both OSD-171-2 and OSD-171-3 consistently. Signage items including MERCS signage removal/installation, hazardous voltage sign replacement, site enquiry signage update, and MERCS#5 on new panel antennas are all addressed.
R026	Make sure outdated/worn-out signage needs to be replaced with new.	PASS	The FC drawing (M8398-G1) Site Signage section explicitly documents replacement of outdated/faded signage including: existing faded hazardous voltage sign on meter box to be replaced with new signage (item 7), existing outdated site enquiry sign on shelter access door to be replaced with new signage (item 6), MERCS#2 signage to be installed on rooftop access door with RFNSA number hard-stamped (item 5), and references OSD-171-2 and OSD-171-3 for correct signage placement standards. These requirements align with the reference image showing replacement of faded Keep Out signs, outdated hazardous voltage signs, and correct placement of signage per OSD171.
R027	Update Electrical Installation and site earthing as per CPS recent remarks. (Only CPS)	PASS	The FC drawing (M8398-G1) includes an 'ELECTRICAL INSTALLATION AND SITE EARTHING' section stating the existing 40A 3-phase supply is insufficient and requires upgrading to 63A 3-phase, with reference to electrical drawings M8398-E1 and E2. Since an electrical upgrade IS required at this site, the drawing correctly documents the upgrade requirement and references the supporting electrical specification and single line diagram sheets, consistent with CPS requirements for the upgrade scenario.
R028	If Electrical upgrade is required, please add E1 & E2 sheets. For E3 sheets add the standard notes in G3 sheets.	PASS	Electrical upgrade is required (G1 sheet states existing 40A supply insufficient and requires upgrade to 63A 3-phase). Both E1 (Electrical Specifications) and E2 (Single Line Diagram) sheets are included in the drawing package as confirmed in the drawing index M8398-00. No E3 sheet is present, so the G3 standard notes requirement for E3 is not applicable.
R029	WHS design risk assessment note included for CPS only. (Not applicable for Servicestream)	PASS	The FC drawing is a ServiceStream drawing (ServiceStream logo and Samina Tabassum listed as ServiceStream distributor on the title block). Per the rule, WHS design risk assessment note is not applicable for ServiceStream drawings. No WHS design risk assessment note is required or expected here.
R030	Update Corrosion Protection note on G1 page. (Only Servicestream site)	FAIL	The G1 page (M8398-G1) of this ServiceStream site does not contain a CORROSION PROTECTION section. The reference image shows a required CORROSION PROTECTION note with CORROSIVITY CATEGORY AS/NZS 2312.2 and PROPOSED CORROSION PROTECTION SYSTEM fields, but no such section is present on the FC drawing's G1 page.
R031	North located and pointed in upper direction	PASS	Both FC drawing pages (M8398-G2 and M8398-G3) contain a North indicator in the upper-left corner showing the arrow pointing upward with 'N' and 'MGA' labels, consistent with the reference image showing a straight north block pointed in the upper direction.
R032	Scale shown and correct (Eg. 1:500/1:1000) Mainly focus atleast two roads for selecting the scale & try to avoid large scale (1:2000 & 1:2500)	PASS	The Overall Site Plan (M8398-G2) is drawn at Scale 1:1500, which is an acceptable standard scale between 1:1000 and 1:2000. The scale successfully covers at least two main roads (Armidale Road, Hazel Glen Drive, Newstead Drive, Painted Hills Road, Kyarra Drive) and the site/meter box location without requiring a large scale of 1:2000 or 1:2500. The detail plan (M8398-G3) uses 1:200, appropriate for a rooftop setout plan.

R033	Street name shown is correct?	PASS	The FC drawing shows the site address as '95 Hazel Glen Drive, Doreen Township' with surrounding streets including Armidale Road, Newstead Drive, Painted Hills Road, Kyarra Drive, and Bathurst Lane. These street names are consistent with the known street layout in Doreen, Victoria as verifiable on Google Maps, confirming the street names shown are correct.
R034	Notes & Legend details included	PASS	Both FC drawing pages include the standard notes (items 1-4) matching the reference requirements. Legend details are present on both pages with correct line types (Existing Property Boundary, Existing Lease Area, Existing Optus U/G Power Supply) corresponding to elements used in the drawings. No O/H power supply lines or fence lines are present in the FC drawings, so those legend entries are not required. Antenna Legend is also correctly included on page G3.
R035	General rule 2 (applicable for all sheets)	PASS	All existing callouts are shown in the FC drawing pages. OSD references are appropriately listed only in the Reference Documents index sheet (M8398-01) and are not embedded as inline callout references within the drawing sheets, consistent with the rule requiring existing callouts to be shown without OSD references within those callouts.
R036	Scale shown and correct (Eg. 1:100) & New equipment shown as BOLD	UNCLEAR	The FC drawing pages provided (M8398-00, M8398-01, M8398-G1) are cover, index, and site specification sheets. M8398-G1 shows only a graphical scale bar but no standard ratio scale (1:100 format) is visible. No detailed plan/elevation viewport pages with equipment are included in the submitted pages to verify scale matching against viewport properties or bold formatting of new equipment. Cannot fully assess compliance without the detailed drawing sheets (G2, G3, etc.).
R037	Compound shown and referenced (as per Indara document for Indara site)	UNCLEAR	The FC drawing pages provided (M8398-00, M8398-01, M8398-G1) are cover/title and site specification sheets only. The compound layout pages (M8398-G2 Overall Site Plan and M8398-G3 Site Layout and Setout Plan) where compound boundaries, dimensions, and lease area should be shown were not included in the submission. Without these pages, it cannot be verified whether the compound is correctly shown and referenced against the Indara document or previous As-Built drawings as required by the rule.
R038	If site owner is INDARA & rooftop site, Earthing notes must be added in G3 sheet.	UNCLEAR	Site is confirmed as INDARA rooftop site (INDARA SITE ID: 3501475, REF LOCATION: 4. ROOFTOP). The G3 sheet (M8398-G3: SITE LAYOUT AND SETOUT PLAN) is listed in the drawing index but the actual G3 sheet page has not been provided in the FC drawing pages for review. Therefore, it cannot be determined whether the required Earthing notes (as shown in the reference image regarding OSD-020 compliance) are included in the G3 sheet.
R039	Show shelter callout	PASS	Shelter callout is present in the Site Specifications drawing (M8398-G1) under the 'EQUIPMENT SHELTER' section, describing 'EXISTING OPTUS PHASE 8 (3m x 2.5m) SANDWICH PANEL EQUIPMENT SHELTER, COLOURED WHITE SUPPORTED ON STEEL FRAME, REFER TO M8398-F1 FOR FURTHER DETAILS.' This follows the standard as-built callout format with shelter type, dimensions, and drawing reference included.
R040	GPS antenna shown: For Nokia region AYGE & Ericsson region GNSS GPS.	UNCLEAR	The FC drawing pages provided (M8398-00 title page, M8398-01 reference documents, M8398-G1 site specifications) do not contain the RRU schedule or GPS antenna model callouts needed to verify whether the correct GPS antenna (Nokia AYGE for Nokia RRUs or Ericsson GNSS GPS for Ericsson RRUs) is specified. The antenna system configuration sheets (M8398-A1, M8398-A2) and physical asset summary table (M8398-A3) required for this check were not included in the submitted pages.

R041	Show removed feeders details & proposed Hybrid cables details. Please check proposed feeders can be possible to accommodate existing cable tray/ladder/underground conduit or internal monopole.	UNCLEAR	The FC drawing pages provided (M8398-00, M8398-01, M8398-G1) are cover/title and site specification sheets only. The RF Plumbing Diagrams (M8398-P1 and M8398-P2), which would contain removed feeder details, proposed hybrid cable details, and cable tray/ladder accommodation information as shown in the reference image, have not been included in the submitted FC drawing pages. Unable to verify compliance without these pages.
R042	G3 sheet - Antenna tagged shown	UNCLEAR	The G3 sheet (Site Layout and Setout Plan) was not included in the FC drawing pages provided for review. Only the cover sheet (M8398-00), reference document sheet (M8398-01), and site specifications sheet (M8398-G1) were submitted. Cannot verify antenna tags (antenna numbers and azimuths) without the actual G3 sheet.
R043	Update EME comments as per OSD-171 document & Form A/B.	UNCLEAR	The G1 site specifications sheet contains EME/site signage comments (items 6 and 7) noting replacement of outdated site enquiry sign and faded hazardous voltage sign, consistent with OSD-171 requirements. However, the reference image specifically highlights that leader lines in correct positions on the G2 Overall Site Plan sheet are critical to this check. The G2 sheet has not been provided in the FC drawing pages, so it cannot be confirmed whether the leader lines are positioned correctly pointing to the meter box and shelter door locations as shown in the reference image.
R044	Roof levels shown (Rooftop sites only)	UNCLEAR	The FC drawing pages provided (M8398-00, M8398-01, M8398-G1) do not show roof level elevations. The Site Specifications sheet (M8398-G1) mentions '8.65m HIGH BUILDING PARAPET WALL' but does not show labeled roof level elevations (RL/AHD values) as required and shown in the reference image. The Site Elevation sheet (M8398-G4) is listed in the drawing register but was not included in the provided pages for review — roof levels may be present on that sheet.
R045	Designated walkway and step overs shown as necessary (Rooftop sites only)	PASS	The site is identified as a rooftop site (REF LOCATION: 4. ROOFTOP). The existing site notes reference 'EXISTING RRU MOUNT CONNECTED TO CONCRETE WALL' and describe access via 'LADDER AND STEP PEGS WITH FALL ARREST SYSTEM PROVIDED ON ROOFTOP & MOUNTING POLE,' indicating this is a concrete rooftop structure. Per the rule, walkways are not mandatory for concrete rooftops. The drawing appropriately references access methods suitable for this concrete rooftop site type.
R046	Safety handrailing / anchor points shown (Rooftop sites only)	UNCLEAR	The FC drawing pages provided only include the cover sheet (M8398-00), reference documents (M8398-01), and site specifications (M8398-G1). No site layout/plan drawings (e.g., G2/G3) are included in the submitted pages to verify if handrailing and anchor points are graphically shown as required. The site specs note mentions a fall arrest system but no handrail or anchor point callouts are visible in the drawings provided for review.
R047	Panel/AAU antenna(s) shown, orientation	UNCLEAR	The FC drawing pages provided (cover sheet M8398-00, reference documents M8398-01, and site specifications M8398-G1) do not include the antenna layout/setout plan or antenna system configuration sheets where panel/AAU antenna azimuths and dimensions would be shown. The reference image flags a dimension discrepancy (antenna model RRVV2VV-6533D-R8 should be 2577×498×197) and azimuth callout issues, but the relevant FC drawing pages (e.g., M8398-G3, M8398-A1/A2) were not provided for comparison.

R048	Maintain 500mm Horizontal, Maintain 70° for passive antennas & 60° AAUs Angular separation. If this separation is not achieved, please highlight to DEs & TL	UNCLEAR	The FC drawing pages provided (M8398-00, M8398-01, M8398-G1) are cover/title sheets, reference document lists, and site specifications only. No antenna layout plan, top-of-structure plan, or antenna mount detail drawings showing horizontal separation dimensions or angular separation between antennas have been included. The relevant sheets (e.g., M8398-G3, M8398-S4 to S7 antenna mount details) needed to verify minimum 500mm horizontal separation and 70°/60° angular separation are not present in the submitted pages. Cannot confirm compliance without these drawings.
R049	Access must be checked for antennas/RRUs. If any access restriction identified, please highlight to DE & CE at early stage. (Refer snip 7 in next sheet for access details)	UNCLEAR	The site specifications (M8398-G1) confirm that ladder and step pegs with fall arrest system are present on the rooftop and mounting pole, meaning climbing area clearance must be checked per the rule. However, the antenna layout plan, elevation drawings, and RRU/antenna mount detail sheets (M8398-G4, S4-S8) were not provided in the FC drawing pages, so it cannot be confirmed whether any RRU or antenna obstructs the climbing area. A full check requires review of the antenna layout plan and elevation drawings to verify no obstructions exist in the climbing path.
R050	Antenna, RRU, Combiner/Filter model & QTY need to match FR/RLM/DPD	UNCLEAR	The FC drawing pages provided (M8398-00 cover sheet, M8398-01 reference documents, M8398-G1 site specifications) do not contain antenna, RRU, or combiner/filter model and quantity callouts. The relevant pages for this check (e.g., site layout/setout plan M8398-G3, antenna system configuration M8398-A1/A2, RRU setout plan M8398-G3-3) were not included in the submitted FC drawing pages. Therefore, it is not possible to verify whether the antenna models/quantities match the reference image (FR/RLM/DPD).
R051	Antenna numbering conforms to OSD70 and RF guidelines - Sector arrow shown correctly for new antenna and existing antennas with ("J" for joint venture and "O" for optus)	UNCLEAR	The antenna system configuration pages (M8398-A1 and M8398-A2) where antenna numbering (e.g., 11-J, 21-J, 31-J) and sector arrows would be displayed were not included in the FC drawing pages provided. Only the title page (M8398-00), reference documents page (M8398-01), and site specifications page (M8398-G1) were submitted for review. Cannot verify antenna numbering convention compliance without the relevant configuration sheets.
R052	New structural design layout (Face mount, headframe from 3rd party or structural H frame, etc) incorporated in this drawing?	UNCLEAR	The site specifications (M8398-G1) indicate that new panel antennas are to be installed on an existing turret on an existing mounting pole, with structural adequacy certified by ALLTELECOS (CERT_M8398 dated 02/04/2025). The drawing package index lists antenna turret mount detail sheets (M8398-S9, S10, S11) and antenna mount details (M8398-S4 to S7), which should show the structural design layout and callouts. However, these structural detail sheets are not included in the FC drawing pages provided for review, making it impossible to confirm whether the turret mount orientation, callouts, and layout are correctly and clearly shown as required by the rule.
R053	Legends and notes shown and read correctly	UNCLEAR	The Antenna Layout Plan sheets (M8398-A1 to A3) which should contain the standard notes (antenna orientations, DRG references, corroded U-bolts) and antenna legend are not included in the FC drawing pages provided for review. Only the title sheet (M8398-00), reference documents sheet (M8398-01), and site specifications sheet (M8398-G1) were provided. Cannot verify whether standard notes, site-specific painting notes, and antenna legend are correctly shown on the antenna layout plan sheets.
R054	Overall structural height EL noted and RL shown at ground level datum point	UNCLEAR	The FC drawing pages provided (cover sheet M8398-00, reference documents M8398-01, and site specifications M8398-G1) do not include the site elevation drawing (M8398-G4) which would show the overall structural height EL and ground level RL datum. The drawing register confirms M8398-G4 exists as 'SITE ELEVATION' but it was not provided for review. Additionally, this is a rooftop site (RFNSA: 3754005) with an 8.65m parapet wall, so the elevation drawing with EL/RL annotations cannot be verified from the pages supplied.

R055	Antenna model name, QTY & Elevation level	UNCLEAR	The FC drawing pages provided (M8398-00, M8398-01, M8398-G1) are the title sheet, reference documents sheet, and site specifications sheet only. The antenna elevation drawing (M8398-G4) and antenna system configuration/physical asset summary sheets (M8398-A1, A2, A3) required to verify antenna model names (RRVV2VV-6533D-R8, AIR3268-B78Y, AVQL NR35), QTY (3 each), and ACL elevation levels (23.1m and 24.1m) against the reference are not included in the submitted FC drawing pages. Unable to confirm compliance without these critical pages.
R056	RRU model, QTY & Elevation level	UNCLEAR	The FC drawing pages provided (M8398-00 cover sheet, M8398-01 reference documents, M8398-G1 site specifications) do not contain the RRU elevation, model, or quantity details needed for this check. The relevant pages (M8398-G3-3 RRU Setout Plan, M8398-G4 Site Elevation, M8398-A1/A2 Antenna System Configuration) were not included in the submission. Cannot verify RRU model (RRU4480/RRU4466), QTY, or elevation levels against the reference image showing EL 23.1m/24.1m heights and highlighted discrepancies.
R057	Feeders model & Qty and feeders route details	UNCLEAR	The FC drawing pages provided only include the cover sheet (M8398-00), reference document sheet (M8398-01), and site specifications (M8398-G1). The critical sheets needed to verify feeder model & qty and feeder route details — specifically the G2 Overall Site Plan and G4 Site Elevation — are not included in the submitted pages. Without these sheets, it is not possible to check feeder cable models, quantities, routing details (cable ladder/tray/conduit), or whether the existing cable route has sufficient capacity for the proposed feeder scope.
R058	Panel/Parabolic antenna heights noted (CL). Vertical clearance dimensions between new and existing antennas mentioned	UNCLEAR	The FC drawing pages provided (M8398-00 cover sheet, M8398-01 reference documents, and M8398-G1 site specifications) do not include the Site Elevation drawing (M8398-G4) where antenna centerline heights (CL) and vertical clearance dimensions between new and existing antennas would typically be shown. The reference image demonstrates the correct format with EL heights noted for each antenna level and vertical separations shown. Without the elevation drawing page being provided, it is not possible to verify compliance with this rule.
R059	Does the headframe, antennas, shelter/ODU and other equipment need painting?	UNCLEAR	The FC drawing pages do not include site photos showing whether existing antennas, mounts, or steelwork are painted. No painting notes are present in the drawing's Notes section. The Equipment Shelter is noted as coloured 'white', but there is no explicit statement or note addressing whether proposed equipment needs to be painted to match existing. Per the rule, site photos should be checked and painting notes added to 'Notes' if required — neither is visible in the provided pages.
R060	Show GPS, shelter & Existing callouts like fence, existing tower callout, other operator shelter callout.	UNCLEAR	The FC drawing pages provided (M8398-00, M8398-01, M8398-G1) do not include the Site Elevation sheet (M8398-G4) or G3 sheet where GPS, shelter, fence, existing tower, and other operator shelter callouts should appear. The relevant elevation drawing required to verify this rule was not supplied for review.
R061	ROOFTOP SITE - Separation Distance between base of antenna and rooftop provided?	UNCLEAR	The FC drawing pages provided (cover sheet M8398-00, reference documents M8398-01, and site specifications M8398-G1) do not include the antenna mount detail or section drawings where the separation distance between the antenna base and rooftop would be shown. The structural antenna mount detail sheets (M8398-S4 through S7) referenced in the drawing register were not provided for review. Cannot confirm presence of minimum 400mm separation dimension without these sheets.

R062	Calculate RF tail length for all RRUs - please consider horizontal & vertical distance for tail calculation	FAIL	The FC drawing does not include any RF tail length calculations for the RRUs. From the elevation drawing (M8398-G4), passive antennas are at EL 15.00m while RRUs range from EL 12.20m down to EL 9.10m, giving vertical distances of 2.8m to 5.9m. RRUs at EL 9.10m have a vertical gap of ~5.9m from the passive antenna, which already exceeds the 5m limit before adding horizontal run and bending radius. The reference image requires explicit annotation of horizontal distance + vertical distance + bending allowance = total RF tail length, and client approval notation if >5m. None of these calculations or annotations are present in the FC drawing pages (G3, G3-2, G3-3, G4).
R063	Check DPD legends are standard or not	UNCLEAR	The FC drawing pages provided (M8398-00, M8398-01, M8398-G1) are the cover sheet, reference documents, and site specifications pages. None of these pages contain a DPD legend. The DPD legend would typically appear on the RF plumbing diagram or antenna system configuration sheets (e.g., M8398-P1, M8398-A1/A2), which were not included in the submitted pages. Unable to verify compliance with the latest DPD legend standard without the relevant drawing pages.
R064	Conjunction notes to refer A1 sheet	UNCLEAR	The FC drawing pages provided (M8398-00, M8398-01, M8398-G1) do not include the RF Plumbing/A1 sheet where the conjunction note 'THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRAWING [SiteID]-A1, A2 & A3' would typically appear. The A1 sheet (M8398-A1) is listed in the drawing index but not provided for review, making it impossible to verify the conjunction note Site ID and drawing number reference.
R065	Shelter type with door swing direction correct	UNCLEAR	The Equipment Shelter Layout Plan (M8398-F1) which would show the door swing direction was not included in the provided FC drawing pages. Only cover sheets (M8398-00, M8398-01) and the Site Specifications (M8398-G1) were provided. Without the shelter floor plan showing the door swing arc, it is not possible to verify whether the door swing direction in the drawing matches the reference photo showing the door opening outward to the right.
R066	Internal rack number to be anticlockwise as enter the shelter, number G000 for ground & R000 for rooftop	UNCLEAR	The Equipment Shelter Layout Plan (M8398-F1) was not included in the provided FC drawing pages, so the internal rack numbering cannot be verified. The site is identified as a rooftop site (REF LOCATION: 4. ROOFTOP on M8398-G1), meaning rack numbers should use R000, R001, etc. format rather than G000, G001. Without the shelter floor plan page, anticlockwise direction and correct prefix cannot be confirmed.
R067	Show all equipment inside the shelter like existing damper, O/H cable tray & fan. Check all internal photos & update the floor plan	UNCLEAR	The Equipment Shelter Layout Plan drawing (M8398-F1) was not included in the FC drawing pages provided for review. Only the cover sheet (M8398-00), reference document sheet (M8398-01), and site specifications (M8398-G1) were provided. Without the actual floor plan page, it is impossible to verify whether the existing damper, O/H cable tray, and fan are correctly shown in the correct locations as required by the rule.
R068	Glandplate sufficient for upgrade works	UNCLEAR	The FC drawing pages provided (M8398-00, M8398-01, and M8398-G1) are title sheet, reference document list, and site specifications pages respectively. None of these pages contain gland plate window photos or visual evidence confirming sufficient space for additional proposed feeders. The reference image shows a gland plate with annotation 'Enough space for proposed feeder', but no equivalent gland plate photo or confirmation annotation is present in any of the submitted FC drawing pages. A dedicated gland plate photo page (likely within the shelter/fitout or site photo section) would be needed to properly assess this rule.

R069	Existing A/C units quantity mentioned including existing stacker frame	UNCLEAR	The FC drawing pages provided are cover sheets and site specifications (M8398-00, M8398-01, M8398-G1) and do not include the equipment shelter layout plan (M8398-F1) where the existing AC unit quantity and stacker frame callout would be shown. The reference image indicates the correct callout should read 'EXISTING 3 STACKER CAGE + L600P AIR CONDITIONERS (2 OFF)' based on site photos. Cannot verify compliance without the F1 shelter layout drawing page.
R070	AC units working conditions	FAIL	The reference image shows 2 OFF split AC units with a water leakage issue captured by the SDV Engineer. None of the FC drawing pages (M8398-00, M8398-01, M8398-G1) contain any notation or callout regarding the AC unit working conditions or the water leakage issue. This issue has not been highlighted in the DFC as required.
R071	Remove Huawei BTS rack	UNCLEAR	The FC drawing pages provided only include the cover sheet (M8398-00), reference documents (M8398-01), and site specifications (M8398-G1). The equipment shelter layout plan (M8398-F1), which would show the Huawei BTS rack removal with RFM and BBU units as seen in the reference photo, has not been included in the pages provided for review. Cannot confirm whether the Huawei BTS rack removal is correctly captured in the drawing scope.
R072	Please make sure use proper BTS equipment based Nokia BBU or Ericsson BBU	UNCLEAR	The FC drawing pages provided (cover sheet M8398-00, reference documents M8398-01, and site specifications M8398-G1) do not contain the BBU model specification. The Equipment Shelter Layout Plan (M8398-F1) which would show the BBU type and the Optus Notes section with BBU callouts are not included in the provided pages. Cannot verify whether proper Nokia or Ericsson BBU is specified without seeing the shelter layout and equipment notes pages.
R073	Existing Krone panel to be upgraded	UNCLEAR	The FC drawing pages provided (M8398-00, M8398-01, M8398-G1) are cover sheet, reference documents, and site specifications pages only. The Equipment Shelter Layout Plan (M8398-F1) where alarm panel details would be shown is not included in the provided pages. No mention of Krone panel upgrade to RJ45 or 6610 PSU alarm panel is found in the visible pages, making it impossible to verify compliance with the Krone-to-RJ45 upgrade rule.
R074	Existing Skinny MER rack & 15-way iDCDP reused - please make sure more than 15-iDCDP are not accommodate.	UNCLEAR	The FC drawing pages provided (cover sheet, reference document list, and site specifications M8398-G1) do not include the Equipment Shelter Layout Plan (M8398-F1) which would show the existing Skinny MER rack and iDCDP configuration. Without the shelter fitout drawing, it is not possible to verify whether a 15-way iDCDP is being reused with the existing Skinny MER rack or if a more-than-15-way iDCDP has been specified.
R075	Existing/New 18-way/22-way iDCDP proposed for new/existing 37RU MER rack	UNCLEAR	The FC drawing pages provided (M8398-00, M8398-01, M8398-G1) do not include the Equipment Shelter Layout Plan (M8398-F1) where iDCDP and MER rack details would typically be shown. Without this page, it cannot be confirmed whether the existing/new 22-way iDCDP is properly documented for the proposed upgrade works on the 37RU MER rack.
R076	Remove Huawei PSU/Invenys and any batteries older than 5 years after the new installation	UNCLEAR	The FC drawing pages provided (cover sheet, reference documents, and site specifications) do not include the electrical specifications (M8398-E1), single line diagram (M8398-E2), or equipment shelter layout plan (M8398-F1) which would contain PSU type details and battery information. No mention of Huawei PSU/Invenys removal or battery age requirements was found in the available pages. The relevant electrical drawings are referenced but not provided for review.

R077	Propose or reuse existing PSU & rectifier based on PDT. For eJV min 18kW are required whereas Optus only site use Min. 16kW PSU. If less than this condition, replace with new one.	UNCLEAR	This is an eJV site (Joint Venture Site ID: JM6015) requiring minimum 18kW PSU from Vertiv. However, the PSU Deployment Summary sheet (PRP-SOL tab) showing PSU model, capacity, rectifier configuration, and brand details is not included in the FC drawing pages provided. Only the cover page (M8398-00), reference documents (M8398-01), and site specifications (M8398-G1) were submitted. The PSU selection and configuration details cannot be verified without the relevant electrical/PSU summary sheet.
R078	Battery type & Model. If an Optus shelter is installed inside a Telstra exchange building, lithium batteries are not allowed. Even if the shelter is installed outside but within 2 m of the Telstra exchange room, lithium batteries are still not allowed	UNCLEAR	The FC drawing pages do not include a PSU/PDT sheet showing the battery model and type (no PRP-SOL or equivalent sheet visible). The site is an Indara rooftop installation (not a Telstra exchange building), so lithium batteries would be permissible. However, the battery model cannot be confirmed from the provided FC drawing pages as the electrical/PDT summary sheet is missing. The shelter location is confirmed as rooftop (non-Telstra exchange), which satisfies the location condition of the rule.
R079	Replace/reuse PSU CB size based on PDT tools - Vertiv PSU min. required 40A 1P CB (3 OFF)	UNCLEAR	The FC drawing pages provided (cover sheet, reference documents, site specifications) do not contain the IDB/MCB CB rating details needed to verify PSU CB sizes. The relevant electrical pages (M8398-E1, E2) or PRP-SOL sheet showing PSU CB ratings are not included. The reference image shows PSU 1 and PSU 2 proposed at 40A but PSU 3 remains at 0, suggesting only 2 OFF 40A CBs are proposed rather than the required 3 OFF 40A 1P CBs for Vertiv PSU.
R080	Mention PDT approved version number. Please make sure use recent PDT tools	FAIL	No PDT approved version number is mentioned anywhere in the FC drawing pages reviewed (M8398-00, M8398-01, M8398-G1). The reference image shows that the drawing should include a statement such as 'OPTUS POWER TOOL V12.7 APPROVED' in the notes/specifications section, but this is absent from the FC drawing.
R081	Ensure the exact AC power supply value is mentioned, and it must align with the PVA, power meter photos, and SLD. Also, confirm if a power upgrade is required	FAIL	The FC drawing (M8398-G1) states the existing AC power supply is 40A 3-Phase and requires an upgrade to 63A. However, the PVA report confirms the Optus supply is 50A 3-Phase (meter SN:218038715, 50A protection/main switch rating), and the PDT AC Power Summary shows existing capacity of 50A is sufficient for the proposed upgrade with no upgrade required. The stated 40A value does not align with the PVA, and the power upgrade statement contradicts both the PVA and PDT findings.
R082	If RRU is internal solution, need to show RACK layout	PASS	The RRUs at site M8398 are externally mounted on the rooftop turret/concrete wall mount (as stated in site specifications: 'NEW RRU'S TO BE INSTALLED ON EXISTING RRU MOUNT CONNECTED TO CONCRETE WALL'). This is an external RRU solution, not an internal shelter/ODU cabinet installation. Therefore, the requirement to include a separate rack layout sheet does not apply. The drawing package includes M8398-F1 (Equipment Shelter Layout Plan) for the shelter fitout, which is appropriate for this configuration.
R083	All text font in ISOCP	PASS	The reference image shows that correct drawings use ISOCP text style (as highlighted in the AutoCAD properties panel). The FC drawing pages (M8398-00, M8398-01, M8398-G1) display text that is consistent with the ISOCP font style - the characteristic single-stroke engineering font appearance is visible throughout all pages in elements such as drawing titles, notes, table content, and annotations. The text appearance matches the ISOCP style shown in the reference image.

R084	All the existing and proposed equipments and texts layers are correct	UNCLEAR	The FC drawing pages shown (M8398-00, M8398-01, and M8398-G1) are title/cover sheets and a site specifications sheet containing only text-based content (drawing registers, reference documents, site notes, and general information). These pages do not contain equipment symbols, callouts, or dimensions that would allow verification of bold vs unbold layer usage for existing vs proposed equipment. The reference image shows a site plan/layout drawing where the distinction between existing (unbold) and proposed (bold) layers can be visually assessed. Without seeing the actual site plan, antenna plan, or shelter layout drawings from this FC package, it is not possible to confirm whether the layer convention is correctly applied throughout the project.
R085	Dimensions and leader line not clash and mentioned correctly	PASS	The FC drawing pages provided (M8398-00, M8398-01, M8398-G1) are administrative/title/specification pages containing no dimension lines or leader lines with arrowheads. There are no callouts, leader lines, or dimensions present on these pages that could clash or have missing arrowheads, so no violations of the dimension and leader line rule are detected.
R086	Spell check done properly	PASS	No spelling errors or duplicate callouts were detected across the three FC drawing pages (M8398-00, M8398-01, M8398-G1). All section headings, notes, and labels are correctly spelled. Technical terms such as 'ALLTELCOS', 'MERCs#', and abbreviations appear consistent and intentional. No duplicate callouts were identified.